Voice Based Stress Analysis and Detection using Machine Learning

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***Abstract*—With a major focus on monitoring depression, this research presents a novel machine learning approach for voice-based stress analysis and detection. Our technique uses advances in artificial intelligence to anticipate the Beck Depression Inventory II (BDI-II) scores based just on speech patterns. We use a multi- pronged approach to accomplish this, which involves extracting mel-frequency cepstral coefficients and spectral low-level descriptors from brief audio clips. These characteristics pick up on subtleties in voice expressions that could point to underlying depression or stress. Additionally, in order to improve our model's capacity to identify patterns linked to depression, we suggest using a feature dynamic history histogram (FDHH) to record temporal changes in the feature space. Regression approaches are used to fuse FDHH and audio features, providing a thorough study of vocal data for accurate BDI-II scale prediction. Our approach, which just looks at voice cues, provides a non-invasive and effective way to analyze stress and identify depression. This approach may find use in mental health monitoring and treatment.**

Keywords—Spectral Features, Mel-Frequency Cepstral Coefficients, Feature Dynamic History Histogram (FDHH), Regression Techniques, Beck Depression Inventory II (BDI-II)

I. INTRODUCTION

Deficits in cognitive control have been connected to mental health conditions including depression. Approximately 25% of working-age persons are impacted by it, leading to notable setbacks and difficulties for the judicial, social, educational, and economic systems. A prevalent mental illness known as depression is characterized by symptoms such as low energy, loss of interest or pleasure, depressing mood, feelings of guilt or low self-worth, interrupted sleep or eating, and difficulty concentrating.

Major depressive disorder (MDD), the most frequent psychiatric illness, poses a serious risk to people's mental health. Depression is the leading cause of disability, accounting for 7.5% of all individuals with disabilities more than 300 million people. A recent study suggests that there is a higher likelihood of MDDs among those with low incomes. It may also have an impact on significant life events like when a person gets married and how educated they are. The majority of those who receive treatment for depression don't fully recover from it.[1] The person is still suffering from the sickness. This could manifest as stomach issues, exhaustion, sleeplessness, excessive sleeping, or loss of vitality.

The most common psychiatric condition, major depressive disorder (MDD), is extremely dangerous for people's mental health. Depression is the primary cause of disability, affecting around 300 million people, or 7.5% of all people with disabilities. According to a recent study, those with low earnings may be more likely to have MDDs. Significant life events like a person's marriage and level of education may also be impacted by it. According to, most people who undergo treatment for depression don't make a full recovery. The illness is still causing the victim pain. This could show up as digestive problems, tiredness, insomnia, oversleeping, or a decrease in energy.[3] Machine learning is an artificial intelligence application that enables a system to learn from its experiences and function autonomously without the need for special creation (Artificial Intelligence). [5] Computer programs that can access data and utilize it to learn on their own are created through machine learning.[2] By utilizing "training data," explicit programming builds a mathematical model that is capable of making predictions or decisions through machine learning. Image mining techniques include the following: uncovering hidden information from photos, connecting picture data, and identifying novel, obscure patterns in images. This area includes datasets, image processing, data mining, and machine learning.

Based on conservative estimates, stress is linked to 50–80% of physical diseases in the medical literature. Without needing to be expressly developed, learn from experiences and advance (Artificial Intelligence). Finding new, hidden patterns in photographs, linking picture data, and extracting hidden information from photos are some of the uses for image mining techniques. Datasets, image processing, data mining, and machine learning are all included in this field.[7] According to conservative estimates, the medical literature links stress to 50–80% of physical issues. Stress is considered to be the primary cause of heart disease. Stress is related to diabetes, ulcers, asthma, migraine headaches, skin disorders, epilepsy, and sexual dysfunction. Each of these disorders has a psychosomatic basis, meaning that mental factors like stress either cause or exacerbate the condition, just like many other conditions do. There are

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three known consequences of stress: Stress can cause subjective symptoms such as concern, guilt, annoyance, and embarrassment. In addition, people could experience fatigue, tension, anxiety, irritability, moodiness or loneliness. Stress's behavioral impacts cause noticeable changes in a person's behavior. Warning signs include an increase in accidents, drug or alcohol usage, inappropriate laughing, inflammatory or violent behavior, extremely agitated feelings, and/or binge eating or drinking.[6-15] Fig .1 Show in Speech Audio Extraction for depression scale recognition.

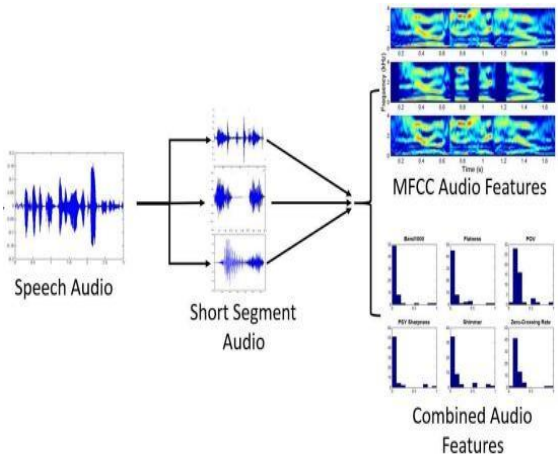

Fig .1 Speech Audio Extraction for depression scale recognition.

II. OBJECTIVE

The main goal of this project is to use machine learning to create a novel method for voice-based stress analysis and detection, with a focus on monitoring depression. The objective is to use chroma characteristics to predict the Beck Depression Inventory II (BDI-II) scales exclusively from vocal expressions. The goal is to create a system that can use spectral low-level descriptors, mel-frequency cepstral coefficients, and chroma features to analyze vocal signals for signs of stress and sadness. This will be accomplished by exploiting improvements in artificial intelligence, chroma characteristics, mel-frequency cepstral coefficients, and spectral low-level descriptors are extracted from brief audio clips in order to capture subtle vocal expressions and musical tone. By adding a feature dynamic history histogram (FDHH), the model's capacity to recognize patterns linked to depression is improved by capturing temporal changes in the feature space. Regression approaches are used to combine FDHH, chroma characteristics, and audio features in order to give a thorough analysis of vocal data and precise BDI-II scale prediction providing a non-invasive, effective method of detecting depression and analyzing stress, with potential uses in mental health monitoring and intervention.

III. LITERATURE SURVEY

A. *IT Professional Stress Detection Using Machine Learning By Image Processing Using Deep Learning*

AUTHORS: A Priya Dharshini, Mrs. P. Jasmine Lois Ebenazer, Mrs. S Cyciliya Pearline Christy and Mrs. S Cephas

The objective is to foster an environment of spontaneity and well-being among employees in a firm in order to boost output. The goal is to gauge an employee's level of stress while they are at work. An employee's captured image will be used as input. Stress level classification is done using KNN classifiers. There are three stages: importing photos with tools for taking pictures, Adjust and examine the picture, examine the image. MTCNNs, or multi-task cascading convolutional networks, are used to identify landmarks in facial photographs. Because the Softmax classifier can differentiate between individuals, it is used. It is used in conjunction with MTCNN to identify faces and provide distinctive facial representations. FaceNet is employed. In general, reliance on particular classifiers will restrict to various scenarios as well as advanced technologies.

B. *Stress Detection Using Machine Learning And Image Processing*

AUTHORS: Mahesh Kanthale, Akshata Shendkar, Payal Kadadhekar, Siddheshwari Rao, Prof. Dheeraj Patil

Utilizing the HAAR Cascade method, faces are detected. A system is utilized to record images, which are then used as a dataset for analysis to display the beat-by-beat frequency of a stressed individual. The KNN method aids in data inspection and classification. One way to determine if someone is under stress is to use supervised learning techniques. However, accuracy attained by human performance is unachievable, and the algorithm might not be exact, providing us with an overhead detection rate.

C. *Artificial Intelligent System For Automatic Depression Level Analysis Through Visual And Vocal Expressions*

AUTHORS: Asim Jan, Hongying Meng, Yona Falinie Binti A. Gaus, and Fan Zhang

Emotion analysis has proven to be a useful method for simulating depressive states. relates to the importance of data for describing various psychiatric conditions. Facial expressions reflect emotions, which the Facial Action Coding System (FACS) decodes. Effective features in a deep space are captured by the feature dynamic history histogram (FDHH). These useful features use the LR and PLS algorithms to map to depression scales. The audio and video are blended to improve the forecast. In conjunction with graphic elements, audio features like Band1000, Flatness, PSY, POV, Shimmer, Sharpness, and ZCR were employed. PCA is used to evaluate depression LR/PLS and minimize dimensionality. FDHH records dynamic patterns, enabling the creation's final visual inspection. Overall, there is little truth in the expression of chord progression and tone quality.

D. *Stress Detection Through Speech Analysis Using Machine Learning*

AUTHORS: Dr. S. Vaikole, S. Mulajkar, A. More, P. Jayaswal, S. Dhas

It talks about utilizing voice analysis and machine learning to detect stress. It suggests a deep learning model that makes use of the Ryerson Audio-Visual Database of Emotional voice and Song (RAVDESS). Using Mel-Frequency Cepstral Coefficients (MFCC), the model

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detects strained emotions with good classification accuracy. The study advises professional cortisol level measurement for more accurate experimental outcomes and raises the possibility of using a multi-modal technique to improve detection accuracy. Along with references and a dataset link, the post offers classification accuracy for various moods and attributes.

E. Automated Stress Detection Using Machine Learning

AUTHORS: Anakha P.S, Anjana S Nair, Aiswarya Devi, Aishwarya Suresh

The necessity for efficient stress detection and management due to the rising incidence of stress in today's culture. It looks at how physiological signals and machine learning can be used to identify stress as well as how stress affects people's mental and physical health. The goal of the suggested stress detection system is to help those who are under stress by identifying them early and offering assistance. The report also includes a comparison of current stress detection systems, along with a suggested system architecture and methodology. The assortment of scholarly articles and investigations pertaining to stress detection through the use of diverse machine learning and deep learning methodologies underscores the possible advantages and drawbacks of each strategy, in addition to the prospects for enhancing stress detection systems in the future.

F. Stress Detection Using Machine Learning And Deep Learning

AUTHORS: Z. Zainudin, S. Hasan, S.M. Shamsuddin and S. Agarwal

Using machine learning techniques to biosignals gathered from sensors in order to identify stress levels. When many classification methods are compared, it is discovered that Decision Tree performs the best. The study highlights that in order to increase accuracy, precision, recall, and F1-score, it is crucial to use an appropriate classifier. The research, which is funded by a number of organizations, recommends more classifiers and 10-fold cross validation be used in future studies.

IV. EXISTING SYSTEM

Using voice expressions, the current system predicts the Beck Depression Inventory II (BDI-II) scales. It does this by employing machine learning approaches for voice-based stress analysis and depression detection. The methodology it employs involves the extraction of mel-frequency cepstral coefficients and spectral low-level descriptors from brief audio clips in order to capture subtleties in vocal expressions that may be symptomatic of sadness or stress.

Drawbacks - The system may miss other crucial indicators like physiological signals or facial expressions because it only employs voice cues for stress analysis and depression diagnosis. The system is made more difficult to develop and maintain by the multifaceted methodology that involves spectral analysis and feature extraction from audio segments. The system's applicability in situations requiring prompt intervention or monitoring is limited because it makes no mention of real-time analysis capabilities. Although the system claims to perform better than current approaches, it's possible that the evaluation is skewed toward particular datasets or scenarios, which could overestimate its usefulness in practical applications. Although the emphasis on vocal signals provides a non-invasive method for analyzing stress and identifying depression, it could undervalue the significance of thorough evaluations that take into account a variety of modalities in order to provide a more precise diagnosis. Fig 2 show in Comparison of Performance: Trained vs New Data set.

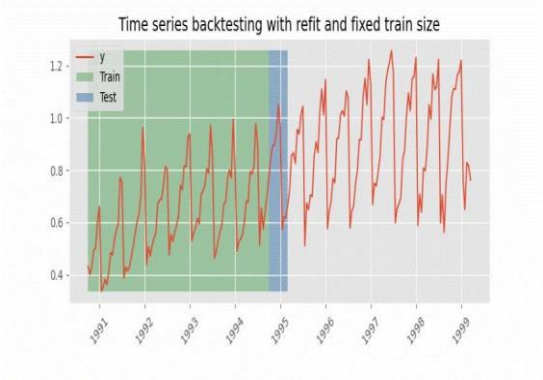
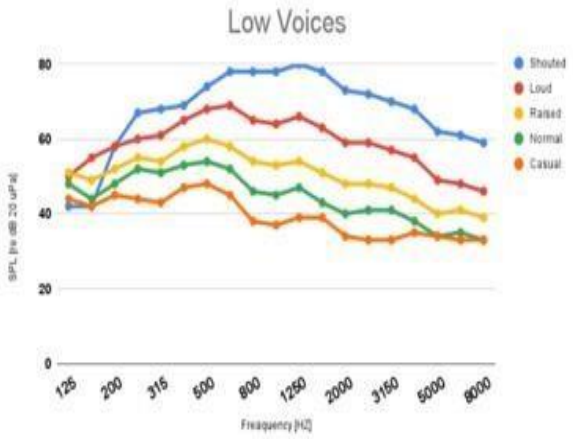


Fig. 2 Comparison of Performance: Trained vs New Data set

V. PROPOSED SYSTEM

Although the work focuses primarily on audio and visual components of analysis, adding chroma features could improve the outcome even further, especially considering the significance of audio quality in the analysis of mood and despair. We pay special attention to stress detection, which is voice-analyzed. Pitch and tone variations can be captured with the use of the chroma feature integration. With ease, we may examine the emotional patterns in speech, giving us a deeper comprehension of the voice's stress markers.

Advantages of the suggested framework - When the chroma feature is integrated, tone information is also recorded in the audio along with the pitch class distribution. makes it possible to recognize melodic elements in speech, which is crucial when a person's speech displays lyrical or tonal patterns connected to emotional states. enhances the model's capacity for distinction. Thus, the incorporation of the chroma feature provides us with a comprehensive framework for identifying and analyzing stress. Fig. 3 Show in Voice Frequency Range. Fig. 4 Show in Block-diagram for stress detection.



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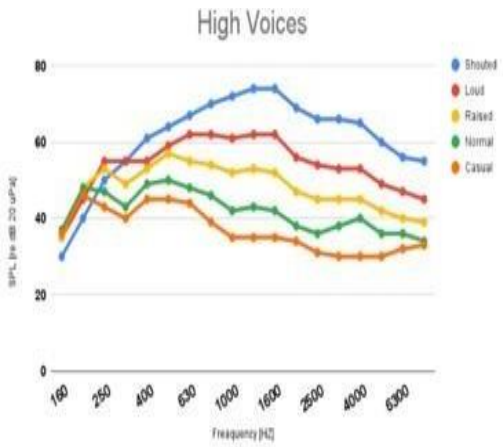

Fig. 3 Voice Frequency Range

VI. METHODOLOGY

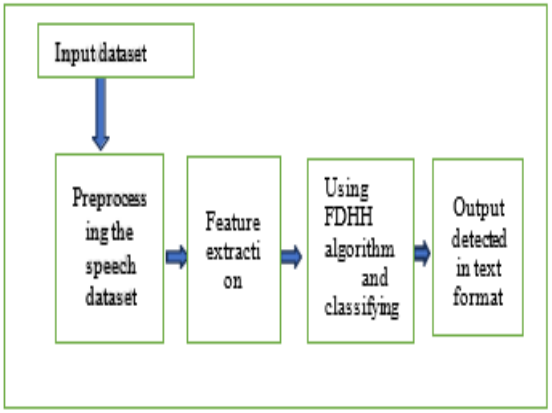

Fig .4 Block-diagram for stress detection.

A. Feature Extraction:

In order to capture vocal expressions, as stated in the current methodology, extract spectral low-level descriptors and mel- frequency cepstral coefficients from brief audio samples. In order to extract pitch and tone information from vocal expressions, incorporate chroma feature extraction with already used feature extraction methods.

B. Preprocessing:

To guarantee that significant chroma features are extracted, preprocess audio data by dividing it into brief frames. To improve the chroma feature representation quality, use signal processing techniques like windowing and normalizing.

C. Chroma Feature Extraction:

To compute chroma vectors from the audio frames, we use chroma feature extraction algorithms like Constant-Q Transform (CQT) or Short-Time Fourier Transform (STFT). Each audio frame should be represented as a chroma vector, which will capture the distribution of the pitch classes, or musical notes, contained in the frame.

D. Feature Fusion:

Mel-frequency cepstral coefficients, other features, and current spectral low-level descriptors should be combined with the retrieved chroma features. Examine how to use feature dynamic history histograms (FDHH) with chroma features to capture temporal changes in the feature space.

E. Model Training And Validation:

Use the fused feature set to train machine learning models, such as ensemble approaches, neural networks, and regression models. To guarantee reliable performance, validate the training models using suitable assessment metrics and cross-validation methods.

F. Performance Evaluation:

Analyze the improved system's performance using benchmark datasets, such as the 2014 Audio/Visual Emotion Challenges dataset. Evaluate the effectiveness of adding chroma characteristics by contrasting the revised system's performance with the current methodology.

G. Interpretation And Analysis:

Examine the roles that chroma characteristics play in stress analysis and depression identification, keeping in mind that these features can record pitch-related changes in vocal emotions. Analyze the findings to comprehend how chroma feature integration affects the Beck Depression Inventory II (BDI-II) scale prediction.

H. Iterative Movement:

The model will be continuously refined by integrating newly discovered label data or by modifying the features. We also consider real-life scenarios and users feedback to enhance the accuracy of the system.

By doing these actions, the system can improve voice-based stress analysis and depression identification by using chroma features, which could lead to an increase in prediction accuracy and dependability.

VII. CONCLUSION

In conclusion, our "Voice-Based Stress Analysis and Detection using Machine Learning" framework has made a substantial breakthrough in the detection and monitoring of stress and depression with the addition of chroma feature extraction. More complex analysis of emotional states is possible thanks to chroma features, which offer insightful information on the pitch and tonal qualities of vocal expressions. Through the use of this data in conjunction with other spectral descriptors and dynamic history histograms, our system is able to provide assessments of mental health that are more precise and thorough.

Moreover, adding chroma characteristics to our machine learning model makes it more successful overall by allowing it to recognize minute patterns and fluctuations in voice data that could be signs of underlying stress or sadness. This development has the potential to enhance mental health monitoring and intervention tactics by enabling earlier identification and more focused assistance for people who are experiencing psychological discomfort.

With everything considered, the addition of chroma feature extraction to our system is a major advancement in the fields of stress analysis and depression identification, opening up new possibilities for raising the standard of assistance and care given to those who require it.

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